

Student Grade Calculator System

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GitHub Statement: A lightweight, modular Python application engineered to process student academic marks, evaluate score metrics, determine letter grades, and generate structured academic results cards. Built with clean functional design, the system isolates data ingestion, logic evaluation, and console presentation into dedicated Python modules.

1. KEY PROJECT FEATURES

- Modular Architecture:** Clean separation of concerns between input handling, calculation business logic, and output formatting.
- Automated Grading Logic:** Assigns letter grades (A+ through F) using precise percentage boundaries.
- Result Evaluation:** Determines pass/fail status using an overall aggregate minimum threshold of 40%.
- Formatted Terminal Output:** Produces a clean, formatted report card directly to standard console output.

2. REPOSITORY STRUCTURE

File Name	Layer	Description
student_input.py	Data Ingestion Layer	Fetches and returns student attributes and subject marks.
grade_calculator.py	Business Logic Layer	Computes aggregate sum, percentage, grade, and pass/fail status.
main.py	Application Orchestration	Coordinates execution flow and prints formatted result card.

3. SOURCE CODE MODULES

3.1 Data Ingestion Module (student_input.py)

student_input.pyData Layer

```
def get_student_data():  
    # Sample student data  
    name = "Rahul Sharma"  
    maths = 85  
    physics = 78  
    python = 92  
    english = 81
```

3. SOURCE CODE MODULES (CONTINUED)

3.2 Business Logic Module (grade_calculator.py)

Contains core mathematical computations and conditional evaluation logic for determining percentage, letter grade assignment, and overall pass/fail status.

grade_calculator.py

Business Logic

```
def calculate_result(maths, physics, python, english):
    # Calculate total marks
    total = maths + physics + python + english

    # Calculate percentage
    percentage = total / 4

    # Calculate grade
    if percentage >= 90:
        grade = "A+"
    elif percentage >= 80:
        grade = "A"
    elif percentage >= 70:
        grade = "B"
    elif percentage >= 60:
        grade = "C"
    elif percentage >= 50:
        grade = "D"
    else:
        grade = "F"

    # Calculate result
    if percentage >= 40:
        result = "PASS"
    else:
        result = "FAIL"

    return total, percentage, grade, result
```

4. GRADE CALCULATION RULES

Percentage Range	Letter Grade	Status Threshold
≥ 90%	A+	PASS
80% – 89%	A	PASS
70% – 79%	B	PASS
60% – 69%	C	PASS
50% – 59%	D	PASS
< 50%	F	FAIL (< 40%)

3. SOURCE CODE MODULES (CONTINUED)

3.3 Execution Entry Point (main.py)

Coordinates execution flow by linking data retrieval, score calculations, and standard stdout report card rendering.

main.py

Entry Point

```
from student_input import get_student_data
from grade_calculator import calculate_result

# Get student data
name, maths, physics, python, english = get_student_data()

# Calculate result
total, percentage, grade, result = calculate_result(
    maths, physics, python, english
)

# Display student result
print("=====")
print("          STUDENT RESULT")
print("=====")

print(f"Name       : {name}")
print(f"Maths      : {maths}")
print(f"Physics    : {physics}")
print(f"Python     : {python}")
print(f"English    : {english}")

print("-----")

print(f"Total       : {total}/400")
print(f"Percentage  : {percentage:.2f}%")
print(f"Grade      : {grade}")
print(f"Result     : {result}")

print("=====")
```

5. EXECUTION CONSOLE OUTPUT

```
=====
          STUDENT RESULT
=====
Name       : Rahul Sharma
Maths      : 85
Physics    : 78
Python     : 92
English    : 81
-----
Total      : 356/400
Percentage : 89.00%
Grade     : A
```